

AMD Ryzen 7 9800X3D

Great processor, sub-par pricing

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MD Ryzen 7 9800X3D is a much more exciting launch than the initial Ryzen 9000

series “Granite Ridge” desktop processors for a very simple reason - 3D V-Cache. Not only do the X3D chips bring all the goodness of the new core microarchitecture but the additional cache memory slapped on (or should we say, Slapped under) the chip. Workloads benefitting from the added cache tend to perform really well with the X3D chips and one of the major workloads that gets a massive boost are video games. Moreover, there's the added power-efficiency of having switched to a more efficient TSMC 4nm process node. So how well does the new AMD Ryzen 7 9800X3D fare in benchmarks? Let's find out.

3D V-CACHE, NOW DOWN-UNDER

No, AMD is not making its chips in Australia. The 3D V-Cache is now placed underneath the CPU dies. With the first implementation of TSMC's CoWoS technology, the additional cache layer was placed on top of the CPU dies. While cache memory does run hot, it's nothing as compared to heat generated by the CPU cores themselves. So flipping the little silicon pancakes makes a lot of sense. Why AMD chose to go with the cache on top or why TSMC might have advised them to do so or why ... it doesn't matter. They flipped the pancake, and it's supposedly running much better. The hotter cores are now close to the heatsink, and therefore, the CPU cooler as well. AMD states that the move has resulted in a massive benefit when looking at the thermal. Also, the 3D V-Cache has up to 46 per cent better thermal resistance. And since the cores

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Performance.....8.1
Features.....6.8
Value for money ... 6.5

are running a little cooler, they can be pushed to run at higher clock speeds. AMD has managed to run the AMD Ryzen 7 9800X3D at a 500 MHz higher base clock and 200 MHz higher boost clock compared to the Ryzen 7 7800X3D. They also state

that multi-core residency has improved a lot and the multi-threaded performance is much closer to the AMD Ryzen 7 9700X which runs at 105W. This is purely because the 9800X3D can better use the power that's provisioned for it.

AMD RYZEN 7 9800X3D SPECIFICATIONS

The AMD Ryzen 7 9800X3D has some specs similar to the Ryzen 7 9700X with the same 8 Cores and 16 Threads, L1 and L2 Caches. The base clock speed goes up by 900 MHz and the boost clock drops by 300 MHz. 3D V-Cache adds 64 MB of L3 Cache on top of the existing 32 MB to give 96 MB of L3 Cache. That's the same amount as the 7800X3D. The TDP has

gone up because there are more components running in the same space and there's also the higher base clock speed leading to more heat on average.

The DRAM speeds are still at 5600 MT/s so not much has changed on the memory front. Intel's recent Core Ultra 200S processors raised the non-OC memory speeds to 6400 MT/s and also added support for CUDIMMs / CSODIMMs. We feel that this seems to have been an easy addition with 6400 MT/s being the memory sweet spot for AMD Zen 5 chips. Aside from these things, the AMD Ryzen 7 9800X3D has nothing different from the recently launched Ryzen 9000 chips.

On the platform front, we've got the X870E motherboards for enthusiasts available right now. The X870, B850, B840 chipset based motherboards will be launched later on.

AMD RYZEN 7 9800X3D TEST RIG

We received a full kit with new 9800X3D processor, SSD, and memory sticks. We didn't use the provided SSD and stuck to our own storage drives. Here's the full specs:

- **CPU:** AMD Ryzen 7 9800X3D
- **Motherboard:** X870E AORUS MASTER
- **RAM:** 2x G.Skill Trident Z5 16 GB
- **SSD:** Kingston Renegade PCIe Gen 4.0 SSD
- **Cooler:** ASUS ROG RYUJIN III 360 ARGB
- **PSU:** Cooler Master MWE Gold 850 V2

AMD RYZEN 7 9800X3D PERFORMANCE

Cores matter a lot in Cinebench and with processors such as the 9950X and the Ultra 9 285K on the charts, the 9800X3D doesn't break into the Top 3 spots in Cinebench 2024 or R23, in both Single-threaded and Multi-threaded runs. The same can be said for Blender and V-Ray which also leverage higher core-counts for better performance. But the tables are turned in WinRAR and